Image-Based Visual Servoing of Micro Aerial Vehicles With Robust Observation and Output Feedback

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***Abstract*—An output feedback control method is proposed in the article with state observers for robust image-based visual servoing (IBVS) regarding micro aerial vehicles (MAVs). Specifically, image features are first selected from the perspective image moments and are projected onto a suitably defined image plane so as to provide decoupled kinematics for translational motion. In order to address the problem of low quality and even unavailability of linear velocity information, the image moment error observer is designed as a foundation based on an image filter, and then the MAV body velocities are estimated by designing a robust observer to obtain translational optical flow of virtual image features. An adaptive law is proposed for the unknown depth information of the visual target and is combined with observers in output feedback controller design to enhance system robustness. Stability analysis shows that all states of the system are uniformly ultimately bounded (UUB) and converge to zero asymptotically. Comparative simulation and experimental results demonstrate the performance of the proposed method.**

***Index Terms*—Depth adaptation, image-based visual servoing (IBVS), robust velocity observer, translational optical flow, micro aerial vehicles (MAVs).**

I. Introduction

NOWADAYS, with the rapid development of micro aerial vehicles (MAVs) in practical applications, researchers have continued to design new models and advanced controllers to enhance the reliability of MAVs in various potential applications. MAVs can be used for aerial vision, such as providing a wide variety of aerial footage, and can perform tasks in dangerous and hard-to-reach areas, such as infrastructure inspections. In addition, MAVs play a significant role in reconnaissance and rescue missions, enabling rapid response to emergencies and providing important information and support [1], [2]. MAV researches generally include developing efficient actuators, using accurate sensors, and designing reliable controllers. The sensor system of a MAV usually includes an inertial measurement unit (IMU) and a global positioning system (GPS). While the IMU system can provide reliable angular velocity and attitude information for the MAV, it cannot accurately estimate translational position and linear velocity [3], [4]. Additionally, GPS provides relatively high-accuracy positioning information but does not operate effectively in indoor and urban canyon environments, and obtaining linear velocity in GPS-free areas is a challenging problem [5], [6].

Much research has been conducted recently aiming at providing environmental and motion information to MAVs, especially in the case of workspaces without GPS signals. In this context, solutions have emerged for MAV systems by sufficiently using visual sensors. The vision sensor is a reliable and low-cost device that can be fused with IMU data to provide effective translational velocity information, and it can also be effectively used for MAV localization relative to the environment [7], [8]. Application of MAV vision systems includes obstacle avoidance [9], [10], attitude estimation [11], [12], simultaneous localization and mapping (SLAM) [13], [14], path planning [15], and so on.

MAV control based on vision systems mainly consists of two patterns: position-based visual servoing (PBVS) and image-based visual servoing (IBVS). PBVS estimates the 3-D Cartesian space information of a target object through the visual sensor and determines MAV motion based on this information [16]. IBVS uses image information of target features to form closed-loop control directly, and it is robust for camera models and without 3-D reconstruction. Compared with PBVS, IBVS does not need to calculate the position of the target under 3-D coordinates; instead, it relies on the projection relationship between the camera and the target for calculation and control. IBVS can better adapt to fast motion in the scene, uncertainty of target motion direction, and distance change [17]. Considering the underactuated traits of quadrotor MAVs and coupling dynamics on the image plane, the IBVS approach for quadrotor MAVs is, moreover, more challenging than that for fully actuated systems [18]. Rotation and translation degrees of freedom need to be decoupled and independently controlled under IBVS frameworks. In [19] and [20], IBVS frameworks are designed by selecting image features with passive traits for decoupling MAV kinematics,

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and then controllers can be concisely designed regarding translation and yaw movement, respectively.

The IBVS control pattern based on the virtual camera has gradually become a research hotspot in recent years. In order to solve the problem that the visual object leaves the camera field of view (FOV), Xie and Lynch [21] propose an input saturation control law to ensure that roll, pitch, and thrust are within specified ranges. In order to handle the visibility constraint, Zhang et al. [22] developed a robust nonlinear model predictive control scheme for MAV IBVS control such that the visual target can stay within camera FOV. Huang et al. [23] designed a visual servoing scheme based on hierarchical homography, which is used for pose extraction to solve the under-actuation problem of the MAV. In order to solve the problem of manually operating for observing visual targets, Shi et al. [24] design an integrated IBVS flight system including target search and IBVS stages.

MAV linear velocities are usually measured by auxiliary devices such as IMUs in the IBVS framework, but measurement accuracy is easily affected by error accumulation and large vibrations. In order to dispose the limitation, Asl and Yoon [25] designed a controller according to the optical flow on the image plane that does not require linear velocity information of the MAV. Fink et al. [26] designed a virtual-camera-based visual servoing approach by using simplified models, where the linear velocity of the MAV is provided by an external motion capture system. Hashim [27] designed a Lie group-based full-state observer for controlling MAV motion, which does not need angular and linear velocity measurements.

Depth information needs to be considered in IBVS controller design, and it is an unknown parameter when the geometric model of the target is not available. Asl et al. [28] proposed an adaptive IBVS scheme to compensate for depth uncertainty, where a linear observer is designed to estimate the velocity of image features regarding the stationary target. Gao et al. [29] proposed an IBVS strategy based on hierarchical images, using a distance sensor to estimate object depth. Zheng et al. [30] designed a visual servoing controller by using the backstepping pattern, and the depth as part of the control gain is set in a certain range, eliminating the need to know the depth information of feature points. It is, however, still challenging to efficiently improve the robustness of MAVs in unknown depth environments and address low quality and even unavailability of linear velocities obtained by IMUs.

In the article, the image moment error observer is first constructed based on an image filter. Then, for MAV body velocity estimation, a robust observer is designed to obtain the translational optical flow of virtual image features. An adaptive control law is proposed for the uncertainty of image feature depth, which makes the controller robust to unknown depth information. Stability analysis proves that system states are bounded and the error signals converge to zero. Comparative simulation and experiment are conducted to verify the performance of the proposed method. The main contribution of the article is to design an output feedback control strategy regarding unknown linear velocity and unknown depth information, where velocity observer and depth adaptive law are constructed for robust MAV control.

II. Kinematics and Dynamics Model

In order to describe kinematics and dynamics of a MAV, two frames are introduced, where the world frame $\mathcal{F}^w$ is fixed on the ground with directions of north-east-down and the body frame $\mathcal{F}^r$ is located at the MAV centroid with directions of forward-bottom-right. The position of $\mathcal{F}^r$ origin under $\mathcal{F}^w$ is denoted as $\boldsymbol{\xi}(t) = [x(t), y(t), z(t)]^T$. Transformation from $\mathcal{F}^w$ to $\mathcal{F}^r$ consists of a rotation matrix $R(t)$ and a translation vector $\boldsymbol{T}(t)$. The rotation matrix $R(t)$ is composed of entities of the Euler angle vector $\boldsymbol{\eta}(t) = [\varphi(t), \theta(t), \psi(t)]^T$, which denote MAV roll, pitch, and yaw, respectively. The Euler angles rotate orderly to form $R(t)$ as follows:

$$\begin{aligned} R &= R(\psi)R(\theta)R(\varphi) \\ &= \begin{bmatrix} c_\theta c_\psi & -c_\varphi s_\psi + c_\psi s_\varphi s_\theta & s_\varphi s_\psi + c_\varphi c_\psi s_\theta \\ c_\theta s_\psi & c_\varphi c_\psi + s_\psi s_\varphi s_\theta & -c_\psi s_\varphi + c_\varphi s_\psi s_\theta \\ -s_\theta & c_\theta s_\varphi & c_\theta c_\varphi \end{bmatrix} \end{aligned} \tag{1}$$

where $s_\varphi = \sin\varphi(t)$, $c_\varphi = \cos\varphi(t)$, $t_\varphi = \tan\varphi(t)$, and so on.

The kinematics and dynamics of the quadrotor MAV can be expressed as follows [24]:

$$\dot{\boldsymbol{\xi}} = R\boldsymbol{v} \tag{2}$$

$$\dot{R} = R\mathrm{sk}(\boldsymbol{w}) \tag{3}$$

$$m\dot{\boldsymbol{v}} = -m\boldsymbol{w} \times \boldsymbol{v} + \boldsymbol{F} \tag{4}$$

$$J\dot{\boldsymbol{w}} = -\boldsymbol{w} \times J\boldsymbol{w} + \boldsymbol{\tau} \tag{5}$$

$$\boldsymbol{F} = -U_1\boldsymbol{e}_3 + mgR^T\boldsymbol{e}_3 \tag{6}$$

where m is MAV mass, $J \in \mathbb{R}^{3\times 3}$ is the constant inertia matrix, g is the gravitational acceleration, $\boldsymbol{v}(t) = [v_x(t), v_y(t), v_z(t)]^T$ and $\boldsymbol{w}(t) = [\omega_x(t), \omega_y(t), \omega_z(t)]^T$ denote MAV linear and angular velocities, respectively, $\boldsymbol{F}(t) \in \mathbb{R}^3$ and $\boldsymbol{\tau}(t) \in \mathbb{R}^3$ represent resulting external force and torque under $\mathcal{F}^r$, respectively, $U_1(t)$ is total thrust, symbol $\mathrm{sk}(\cdot)$ denotes the skew-symmetric matrix, and $\boldsymbol{e}_3 = [0, 0, 1]^T$ denotes a unit vector.

The Jacobian matrix between the variation of Euler angle vector $\boldsymbol{\eta}(t)$ and angular velocity vector $\boldsymbol{w}(t)$ is given by

$$\begin{bmatrix} \dot{\varphi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & s_\varphi t_\theta & c_\varphi t_\theta \\ 0 & c_\varphi & -s_\varphi \\ 0 & s_\varphi/c_\theta & c_\varphi/c_\theta \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}. \tag{7}$$

III. Image Feature Dynamics

Perspective projection patterns are widely discussed to conduct visual servoing of MAVs, where appropriately defined image moments are mainly adopted as visual features. By using MAV roll and pitch angles, image data are reprojected to a virtual image plane, which is parallel to the object. The image moments can avoid the influence of single-point features from environmental noise and dynamic interference and provide stable image feature measurement. The mathematical form of image moments can be directly used in control design. Based on image moments selected in the virtual image plane, image-based full-actuation controllers can be designed for the MAV to maintain good behavior in Cartesian space.

The horizontal virtual camera model is introduced with the same optical center and parameters as the downward camera, whose frame is denoted as $\mathcal{F}^v$. Considering there are N stationary point features located in a horizontal plane in

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$\mathcal{F}^w$, it can have the same depth regarding $\mathcal{F}^v$. In order to achieve MAV translational control, an image moment vector $\boldsymbol{q}(t) = [q_x(t), q_y(t), q_z(t)]^T$ is selected and computed as follows [24]:

$$q_x = q_z \frac{{}^v u_g}{\lambda}, \quad q_y = q_z \frac{{}^v n_g}{\lambda}, \; q_z = \sqrt{\frac{a^*}{a}} \tag{8}$$

where

$${}^v u_g = \frac{1}{N}\sum_{k=1}^{N} {}^v u_k, \quad {}^v n_g = \frac{1}{N}\sum_{k=1}^{N} {}^v n_k, \; a = {}^v\mu_{20} + {}^v\mu_{02} \tag{9}$$

$${}^v\mu_{ij} = \sum_{k=1}^{N} \left({}^v u_k - {}^v u_g\right)^i \left({}^v n_k - {}^v n_g\right)^j \tag{10}$$

where ${}^v u_k(t)$ and ${}^v n_k(t)$ are the first and second components of the coordinate of the kth point in the virtual image plane, respectively, ${}^v u_g(t)$ and ${}^v n_g(t)$ are the coordinates of the target centroid in the virtual image plane, and ${}^v\mu_{ij}(t)$ corresponds to the virtual pixel distance between the center of visual feature points and the target center. In (10), λ is camera focal length, and a^* is the desired value of $a(t)$ when the MAV reaches the desired pose.

Dynamics of $\boldsymbol{q}(t)$ on the virtual image plane can be obtained as [24]

$$\dot{\boldsymbol{q}} = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{q} - \frac{1}{z^*}{}^v\boldsymbol{v} \tag{11}$$

where z^* denotes constant depth of the target plane to the camera center when the MAV is at the desired pose. The parameter z^* is unknown when the target model is not available, and the proposed controller needs to handle the uncertainty. Only an ideal image of the target is available for general IBVS tasks.

The image feature $q_\psi(t)$ for controlling the MAV yaw motion is defined as follows:

$$q_\psi = \frac{1}{2}\arctan\frac{2{}^v\mu_{11}}{{}^v\mu_{20} - {}^v\mu_{02}} \tag{12}$$

and the time derivative of $q_\psi(t)$ is

$$\dot{q}_\psi = -\dot{\psi}. \tag{13}$$

Formula (11) implies that decoupled and linear relationship exists between virtual image space and Cartesian coordinates in MAV translational motion; thus, only translation and yaw motion of the MAV needs to be controlled to complete IBVS tasks.

IV. Observer and Controller Design

The control task is to adjust the attitude and position of the MAV according to the actual image moment and the desired image moment while realizing the output feedback control based on the image moment error. For controller design, it is affirmed that the camera frame $\mathcal{F}^c$ coincides with the MAV body frame $\mathcal{F}^r$, and only constant transformation between the two frames is required if they do not coincide. The desired image moment vector is first defined as

$$\boldsymbol{q}_d = \left[q_x^d, q_y^d, q_z^d\right]^T = [0, 0, 1]^T \tag{14}$$

where the value $[0, 0, 1]^T$ means that the visual target is in the virtual image plane center, and the MAV is stabilized at a particular height over the target. The image moment error vector $\boldsymbol{e}(t)$ is defined to control MAV translational motion as follows:

$$\boldsymbol{e} = \boldsymbol{q} - \boldsymbol{q}_d. \tag{15}$$

Taking the time derivative of $\boldsymbol{e}(t)$ and using (11), one obtains

$$\dot{\boldsymbol{e}} = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{e} - \frac{1}{z^*}{}^v\boldsymbol{v}. \tag{16}$$

In order to derive translational dynamics of image features regarding the virtual image plane, MAV translational dynamics (4) is re-expressed under $\mathcal{F}^v$ as

$${}^v\dot{\boldsymbol{v}} = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3){}^v\boldsymbol{v} - \boldsymbol{f} \tag{17}$$

where ${}^v\boldsymbol{v}(t) = R_{\varphi\theta}(t)\boldsymbol{v}(t)$ is the MAV linear velocity under $\mathcal{F}^v$, $\boldsymbol{f}(t)$ is the resultant external force acting on the MAV expressed under $\mathcal{F}^v$. It defines $\boldsymbol{f}(t) = [f_x, f_y, f_z]^T = -R_{\varphi\theta}\boldsymbol{F}(t)$ and $R_{\varphi\theta}(t)$ is expressed as $R_{\varphi\theta}(t) = R_\theta(t)R_\varphi(t)$.

In order to design the IBVS controller regarding dynamics (16) and (17), measurement for the velocity ${}^v\boldsymbol{v}(t)$ is required. In this regard, estimated translational optical flow is used to calculate velocity information, which makes it possible to take advantage of the passivity of image error dynamics in IBVS controller design.

Translational optical flow is directly related to linear velocities of camera motion, facilitating the introduction of energy flow in controller design. By mapping optical flow to image errors, the controller can address translational image errors separately.

By differentiating (16) and then substituting (17), system dynamics containing MAV translation and image feature can be presented as follows:

$$\ddot{\boldsymbol{e}} + M_1\dot{\boldsymbol{e}} + M_2\boldsymbol{e} = \frac{1}{z^*}\boldsymbol{f} \tag{18}$$

where $M_1 = 2\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)$ and $M_2 = \mathrm{sk}(\ddot{\psi}\boldsymbol{e}_3) + \mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)$.

In order to facilitate controller and observer design, a variable vector ${}^v\boldsymbol{v}_s(t)$ associated with the linear velocity is defined as

$${}^v\boldsymbol{v}_s = \rho{}^v\boldsymbol{v} \tag{19}$$

where it denotes $\rho = 1/z^*$. It is noted that symbol ${}^v\boldsymbol{v}_s(t)$ is translational optical flow on the virtual image plane, and it is also a scaled linear velocity. Then, (16) is updated by (19), and dynamics of translational optical flow ${}^v\boldsymbol{v}_s(t)$ is obtained by (17) and (19) as follows, respectively,

$$\dot{\boldsymbol{e}} = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{e} - {}^v\boldsymbol{v}_s \tag{20}$$

$${}^v\dot{\boldsymbol{v}}_s = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3){}^v\boldsymbol{v}_s - \rho\boldsymbol{f}. \tag{21}$$

The following auxiliary variables are, moreover, constructed for controller design:

$$\boldsymbol{\zeta} = \boldsymbol{e} - \hat{\boldsymbol{e}} \tag{22}$$

$$\boldsymbol{\varepsilon} = \dot{\boldsymbol{e}} + \boldsymbol{e}_1 \tag{23}$$

$$\boldsymbol{s} = {}^v\hat{\boldsymbol{v}}_s - \boldsymbol{e}_1 + \dot{\boldsymbol{e}} \tag{24}$$

$$\boldsymbol{r} = -\dot{\boldsymbol{\zeta}} - \boldsymbol{\zeta}_1 + 3\dot{\boldsymbol{e}} \tag{25}$$

where $\hat{\boldsymbol{e}}(t)$ and $\hat{\boldsymbol{v}}_s(t)$ are estimates for $\boldsymbol{e}(t)$ and ${}^v\boldsymbol{v}_s(t)$, respectively, $\boldsymbol{e}_1(t) = k_1\boldsymbol{e}(t)$, $\boldsymbol{\zeta}_1(t) = k_1\boldsymbol{\zeta}(t)$, and k_1 is a positive constant gain.

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It is noted that the auxiliary vector $\boldsymbol{\zeta}(t)$ relates to image moment estimation errors regarding translational motion and $\boldsymbol{\varepsilon}(t)$ denotes a filter for the image moment error vector. The vector $\boldsymbol{s}(t)$ includes translational optical flow and image moment error with its derivative. The vector $\boldsymbol{r}(t)$ is designed for $\dot{\boldsymbol{\zeta}}(t)$ convergence analysis, and it includes image moment error derivative, image moment estimation error. In (24) and (25), the proportional coefficients of $\dot{\boldsymbol{e}}(t)$ are determined by the method of indeterminate coefficients regarding following Lyapunov derivative. With the constructed auxiliary variables, velocity observer and depth adaptive law can be incorporated into controller design, and then the MAV thrust can be calculated.

A nonlinear observer is designed to estimate $\boldsymbol{e}(t)$, as shown in the first expression of the following formula. A nonlinear observer is, moreover, developed to estimate ${}^{v}\boldsymbol{v}_s(t)$, as shown in the second and third expressions of the following formula:

$$\begin{cases} \dot{\hat{\boldsymbol{e}}} = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{e} - {}^{v}\hat{\boldsymbol{v}}_{\mathbf{s}} - \varepsilon \\ {}^{v}\hat{\boldsymbol{v}}_{\mathbf{s}} = {}^{v}\boldsymbol{v}_0 - k_2\boldsymbol{\zeta} \\ {}^{v}\dot{\boldsymbol{v}}_0 = -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)({}^{v}\hat{\boldsymbol{v}}_{\mathbf{s}} + k_1\boldsymbol{\zeta} + \varepsilon) - k_1k_2\boldsymbol{\zeta} \end{cases} \tag{26}$$

where k_2 is a positive constant. Based on (20) and (26), we have derivative for the auxiliary vector $\boldsymbol{\zeta}(t)$ as

$$\dot{\boldsymbol{\zeta}} = \dot{\boldsymbol{e}} - \dot{\hat{\boldsymbol{e}}} = {}^{v}\hat{\boldsymbol{v}}_s - {}^{v}\boldsymbol{v}_s + \boldsymbol{\varepsilon}. \tag{27}$$

Considering (26) and (27), the time derivative of ${}^{v}\hat{\boldsymbol{v}}_s(t)$ is

$$\begin{aligned} {}^{v}\dot{\hat{\boldsymbol{v}}}_{\mathbf{s}} &= {}^{v}\dot{\boldsymbol{v}}_0 - k_2\dot{\boldsymbol{\zeta}} \\ &= -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)({}^{v}\hat{\boldsymbol{v}}_s + k_1\boldsymbol{\zeta} + \boldsymbol{\varepsilon}) + k_2\boldsymbol{r} - 3k_2\boldsymbol{\varepsilon} + 3k_1k_2\boldsymbol{e}. \end{aligned} \tag{28}$$

By differentiating (23) and substituting (18), the time derivative of $\boldsymbol{\varepsilon}(t)$ is obtained as

$$\begin{aligned} \dot{\boldsymbol{\varepsilon}} &= \ddot{\boldsymbol{e}} + k_1\dot{\boldsymbol{e}} \\ &= -M_1\boldsymbol{\varepsilon} + k_1M_1\boldsymbol{e} - M_2\boldsymbol{e} + \rho\boldsymbol{f} + k_1\boldsymbol{\varepsilon} - k_1^2\boldsymbol{e}. \end{aligned} \tag{29}$$

Taking the time derivative of $\boldsymbol{s}(t)$ given by (24), and substituting (18), (20), and (21), it can get

$$\begin{aligned} \dot{\boldsymbol{s}} &= {}^{v}\dot{\hat{\boldsymbol{v}}}_{\mathbf{s}} - k_1\dot{\boldsymbol{e}} + \ddot{\boldsymbol{e}} \\ &= -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{s} - 2\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{e}_1 - \mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{\zeta}_1 + k_2\boldsymbol{r} - 3k_2\boldsymbol{\varepsilon} \\ &\quad + 3k_2\boldsymbol{e}_1 - k_1\boldsymbol{\varepsilon} + k_1\boldsymbol{e}_1 - M_1\boldsymbol{\varepsilon} + M_1\boldsymbol{e}_1 - M_2\frac{\boldsymbol{e}_1}{k_1} + \rho\boldsymbol{f}. \end{aligned} \tag{30}$$

Using (18), (25)–(27), time derivative of $\boldsymbol{r}(t)$ in (25) is

$$\begin{aligned} \dot{\boldsymbol{r}} &= -\ddot{\boldsymbol{\zeta}} - k_1\dot{\boldsymbol{\zeta}} + 3\ddot{\boldsymbol{e}} \\ &= -\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{r} + 3\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{\varepsilon} - 3\mathrm{sk}(\dot{\psi}\boldsymbol{e}_3)\boldsymbol{e}_1 - k_2\boldsymbol{r} \\ &\quad + 3k_2\boldsymbol{\varepsilon} - 3k_2\boldsymbol{e}_1 + k_1\boldsymbol{r} + k_1\boldsymbol{\zeta}_1 - 4k_1\boldsymbol{\varepsilon} + 4k_1\boldsymbol{e}_1 \\ &\quad - 2M_1\boldsymbol{\varepsilon} + 2M_1\boldsymbol{e}_1 - 2M_2\frac{\boldsymbol{e}_1}{k_1} + \rho\boldsymbol{f}. \end{aligned} \tag{31}$$

It sets that $\|M_1\| \le \mu_1$ and $\|M_2\| \le \mu_2$, where μ_1 and μ_2 are positive constants. It sets that $\rho_{\min}$ and $\rho_{\max}$ are the minimum and maximum values of ρ, respectively.

The estimate of z^* is defined as $\hat{z}^*(t)$, and is updated with the designed adaptive law as

$$\dot{\hat{z}}^* = k_3\left(\boldsymbol{\varepsilon}^T + \boldsymbol{s}^T + \boldsymbol{r}^T\right)\left(M_1\boldsymbol{e}_1 - M_2\frac{\boldsymbol{e}_1}{k_1}\right) \tag{32}$$

where k_3 is a positive constant. It defines the depth estimation error $\tilde{z}^*(t) = z^* - \hat{z}^*(t)$.

Based on the above analysis, the following IBVS controller is proposed with output feedback:

$$\boldsymbol{f} = -k_r(\boldsymbol{r} - \boldsymbol{s} + \boldsymbol{\varepsilon}) - \hat{z}^*M_1\boldsymbol{e}_1 + \frac{\hat{z}^*}{k_1}M_2\boldsymbol{e}_1. \tag{33}$$

The designed controller (33), moreover, does not need the desired depth z^*, which is replaced by $\hat{z}^*(t)$ that is updated by (32); thus, the controller is robust to unknown image depth information, without using a certain value to replace z^*.

The controller and observer gains k_r, k_1, and k_2 are, moreover, selected to fulfill the following conditions:

$$\begin{aligned} &k_r > \max\left\{\frac{1}{\rho_{\min}}\left(k_1 - \frac{3\mu_1}{8} - \frac{\mu_2}{4k_1}\right), \frac{1}{\rho_{\min}}\left(\frac{k_1}{3} - \frac{\mu_1}{6}\right)\right\} \\ &k_r < \frac{1}{\rho_{\min}}\left(\frac{k_2}{2} - \frac{5\mu_1}{4}\right) \\ &k_2 < \frac{1}{1 - 3k_1}\left(5k_r\rho_{\max} + \mu_1 + \frac{\mu_2}{k_1} - 3k_1\right). \end{aligned} \tag{34}$$

Theorem 1: Consider the dynamic system defined by (18), (20), and (21), and it has states $\boldsymbol{e}(t), \boldsymbol{\varepsilon}(t), \boldsymbol{\zeta}(t), \boldsymbol{s}(t), \boldsymbol{r}(t)$ and input $\boldsymbol{f}(t)$. By the designed controller (33) and adaptive law (32) with parameters (34), the system states $\boldsymbol{e}(t), \boldsymbol{\varepsilon}(t), \boldsymbol{\zeta}(t)$, $\boldsymbol{s}(t), \boldsymbol{r}(t)$ are uniformly ultimately bounded (UUB) and the depth error $\tilde{z}^*(t)$ is bounded, and the states $\boldsymbol{e}(t), \boldsymbol{\varepsilon}(t), \boldsymbol{\zeta}(t), \boldsymbol{s}(t)$, $\boldsymbol{r}(t)$ converge asymptotically to zero.

Proof: With definition, the convergence of the states $\boldsymbol{e}_1(t)$ and $\boldsymbol{\zeta}_1(t)$ to zero is equivalent to the convergence of the states $\boldsymbol{e}(t)$ and $\boldsymbol{\zeta}(t)$. In order to prove the stability of the system, we consider the following Lyapunov function:

$$\begin{aligned} V &= \frac{1}{2k_1}\rho k_r\boldsymbol{e}_1^T\boldsymbol{e}_1 + \frac{1}{2}\boldsymbol{\varepsilon}^T\boldsymbol{\varepsilon} + \frac{1}{2k_1}\rho k_r\boldsymbol{\zeta}_1^T\boldsymbol{\zeta}_1 \\ &\quad + \frac{1}{2}\boldsymbol{s}^T\boldsymbol{s} + \frac{1}{2}\boldsymbol{r}^T\boldsymbol{r} + \frac{1}{2k_3z^*}\tilde{z}^*\tilde{z}^* \end{aligned} \tag{35}$$

where the time derivative of $\tilde{z}^*(t)$ is

$$\dot{\tilde{z}}^* = -\dot{\hat{z}}^*. \tag{36}$$

Taking the time derivative of (35), $\dot{V}(t)$ can be obtained as follows:

$$\begin{aligned} \dot{V} &= \frac{1}{k_1}\rho k_r\boldsymbol{e}_1^T\dot{\boldsymbol{e}}_1 + \boldsymbol{\varepsilon}^T\dot{\boldsymbol{\varepsilon}} + \frac{1}{k_1}\rho k_r\boldsymbol{\zeta}_1^T\dot{\boldsymbol{\zeta}}_1 \\ &\quad + \boldsymbol{s}^T\dot{\boldsymbol{s}} + \boldsymbol{r}^T\dot{\boldsymbol{r}} + \frac{1}{k_3z^*}\tilde{z}^*\dot{\tilde{z}}^*. \end{aligned} \tag{37}$$

Then, substituting (23), (25), (29), and (36) into (37), it can get

$$\begin{aligned} \dot{V} &= \rho k_r\boldsymbol{e}_1^T(\boldsymbol{\varepsilon} - \boldsymbol{e}_1) \\ &\quad + \boldsymbol{\varepsilon}^T\left(-M_1\boldsymbol{\varepsilon} + M_1\boldsymbol{e}_1 - M_2\frac{\boldsymbol{e}_1}{k_1} + \rho\boldsymbol{f} + k_1\boldsymbol{\varepsilon} - k_1\boldsymbol{e}_1\right) \\ &\quad + \rho k_r\boldsymbol{\zeta}_1^T\left(-\boldsymbol{r} - \boldsymbol{\zeta}_1 + 3\boldsymbol{\varepsilon} - 3\boldsymbol{e}_1\right) + \boldsymbol{s}^T\dot{\boldsymbol{s}} + \boldsymbol{r}^T\dot{\boldsymbol{r}} - \frac{\tilde{z}^*}{k_3z^*}\dot{\hat{z}}^*. \end{aligned} \tag{38}$$

Combing $\dot{\boldsymbol{s}}(t)$ result in (30) and $\dot{\boldsymbol{r}}(t)$ result in (31), and substituting control law (33) and adaptive

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updating law (32) into (38), it can obtain $\dot{V}(t)$ as

$$\begin{aligned}\dot{V} = {} & \boldsymbol{e}_1^T(\rho k_r \boldsymbol{\varepsilon} - \rho k_r \boldsymbol{e}_1) \\ & + \boldsymbol{\varepsilon}^T((k_1 - \rho k_r)\boldsymbol{\varepsilon} - k_1 \boldsymbol{e}_1 - \rho k_r \boldsymbol{r} + \rho k_r \boldsymbol{s}) \\ & + \boldsymbol{\zeta}_1^T\left(-\rho k_r \boldsymbol{\zeta}_1 - \rho k_r \boldsymbol{r} + 3\rho k_r \boldsymbol{\varepsilon} - 3\rho k_r \boldsymbol{e}_1\right) \\ & + \boldsymbol{s}^T\left(\rho k_r \boldsymbol{s} - 0.5 M_1 \boldsymbol{\zeta}_1 + (k_1 + 3k_2 - M_1)\boldsymbol{e}_1\right. \\ & \qquad \left. - (k_1 + 3k_2 + M_1 + \rho k_r)\boldsymbol{\varepsilon} + (k_2 - \rho k_r)\boldsymbol{r}\right) \\ & + \boldsymbol{r}^T\left(\rho k_r \boldsymbol{s} + k_1 \boldsymbol{\zeta}_1 - (k_2 - k_1 + \rho k_r)\boldsymbol{r}\right. \\ & \qquad + (3k_2 - 4k_1 - 0.5M_1 - \rho k_r)\boldsymbol{\varepsilon} \\ & \qquad \left. - \left(3k_2 - 4k_1 + 0.5M_1 + M_2 \frac{1}{k_1}\right)\boldsymbol{e}_1\right). \end{aligned} \tag{39}$$

Using the bounds μ_1 and μ_2 defined by the norm of the matrices M_1 and M_2, and using Young's inequality, $\dot{V}(t)$ can be bounded as follows:

$$\begin{aligned}\dot{V} \le {} & -A_p\|\boldsymbol{e}_1\|^2 - B_p\|\boldsymbol{\varepsilon}\|^2 - C_p\|\boldsymbol{\zeta}_1\|^2 - D_p\|\boldsymbol{s}\|^2 \\ & - E_p\|\boldsymbol{r}\|^2 \end{aligned} \tag{40}$$

where the parameters are denoted as

$$\begin{aligned} A_p &= 2\rho k_r + \frac{3\mu_1}{4} + \frac{\mu_2}{2k_1} - 2k_1, \quad B_p = \frac{3\mu_1}{4} + 2k_1 \\ C_p &= \frac{3\rho k_r}{2} + \frac{\mu_1}{4} - \frac{k_1}{2}, \quad D_p = \frac{5\mu_1}{4} - \frac{k_2}{2} - \rho k_r \\ E_p &= \frac{5\rho k_r}{2} + \frac{k_2}{2} + \frac{\mu_1}{2} + \frac{\mu_2}{2k_1} - \frac{3k_1}{2}. \end{aligned} \tag{41}$$

In (40), by adjusting the coefficient of $\tilde{z}^*(t)$ to zero, the effect of the depth estimation error on system stability can be avoided.

Then, a vector is defined as $\boldsymbol{\Omega}(t) = [\boldsymbol{e}_1^T(t), \boldsymbol{\varepsilon}^T(t), \boldsymbol{\zeta}_1^T(t), \boldsymbol{s}^T(t), \boldsymbol{r}^T(t)]^T$, and gains k_r, k_1, and k_2 are selected by satisfying the constraints of (34); therefore, (40) can be reorganized as

$$\dot{V} \le -\beta\|\boldsymbol{\Omega}\|^2 \le 0 \tag{42}$$

where $\beta = \min\{A_p, B_p, C_p, D_p, E_p\}$, and (42) shows that $\boldsymbol{e}(t)$, $\boldsymbol{\varepsilon}(t), \boldsymbol{\zeta}(t), \boldsymbol{s}(t), \boldsymbol{r}(t), \tilde{z}^*(t) \in \mathcal{L}_\infty$. Since $\tilde{z}^*(t) \in \mathcal{L}_\infty$, it can obtain that $\tilde{z}^*(t)$ is bounded.

Since $\boldsymbol{e}(t)$ and $\boldsymbol{\varepsilon}(t)$ are bounded, one gets $\dot{\boldsymbol{e}}(t) \in \mathcal{L}_\infty$ by combining (23). Then, combining $\boldsymbol{\zeta}(t), \boldsymbol{r}(t) \in \mathcal{L}_\infty$, it has $\dot{\boldsymbol{\zeta}}(t) \in \mathcal{L}_\infty$. Since $\boldsymbol{\varepsilon}(t), \boldsymbol{s}(t), \boldsymbol{r}(t) \in \mathcal{L}_\infty$, after substituting (33) into (29)–(31), it can obtain that $\dot{\boldsymbol{\varepsilon}}(t) \in \mathcal{L}_\infty$, $\dot{\boldsymbol{s}}(t) \in \mathcal{L}_\infty$, and $\dot{\boldsymbol{r}}(t) \in \mathcal{L}_\infty$, respectively.

Since $\dot{\boldsymbol{e}}(t), \dot{\boldsymbol{\varepsilon}}(t), \dot{\boldsymbol{\zeta}}(t), \dot{\boldsymbol{s}}(t), \dot{\boldsymbol{r}}(t) \in \mathcal{L}_\infty$, one obtains $\dot{\boldsymbol{\Omega}}(t) \in \mathcal{L}_\infty$, and thus $\boldsymbol{\Omega}(t)$ is uniformly continuous. The following relationship can, therefore, be obtained with (42):

$$\lim_{t\to\infty} \beta \int_{t_0}^{t} \boldsymbol{\Omega}^T \boldsymbol{\Omega} d\tau \le V(\infty) - V(t_0) < \infty. \tag{43}$$

Considering (43) and applying Barbalat's lemma, one gets $\boldsymbol{e}(t), \boldsymbol{\varepsilon}(t), \boldsymbol{\zeta}(t), \boldsymbol{s}(t), \boldsymbol{r}(t)$ converge to zero asymptotically

$$\lim_{t\to\infty} \boldsymbol{e}, \boldsymbol{\varepsilon}, \boldsymbol{\zeta}, \boldsymbol{s}, \boldsymbol{r} = \boldsymbol{0}. \tag{44}$$

Since $\dot{\boldsymbol{e}}(t), \dot{\boldsymbol{\varepsilon}}(t), \dot{\boldsymbol{\zeta}}(t), \dot{\boldsymbol{s}}(t), \dot{\boldsymbol{r}}(t) \in \mathcal{L}_\infty$, we can obtain that $\boldsymbol{e}(t)$, $\boldsymbol{\varepsilon}(t), \boldsymbol{\zeta}(t), \boldsymbol{s}(t), \boldsymbol{r}(t)$ are UUB.

According to (22) and (44), it has

$$\lim_{t\to\infty} \hat{\boldsymbol{e}} = \lim_{t\to\infty} \boldsymbol{e}. \tag{45}$$

Using (23) and (44), one obtains

$$\lim_{t\to\infty} \dot{\boldsymbol{e}} = \boldsymbol{0}. \tag{46}$$

Then, according to (25), (44), and (46), it can obtain that

$$\lim_{t\to\infty} \dot{\boldsymbol{\zeta}} = \boldsymbol{0}. \tag{47}$$

Thus, combing (44) and (47), it has

$$\lim_{t\to\infty} {}^v\hat{\boldsymbol{v}}_s = \lim_{t\to\infty} {}^v\boldsymbol{v}_s. \tag{48}$$

It is, therefore, known that ${}^v\hat{\boldsymbol{v}}_s(t)$ converges to ${}^v\boldsymbol{v}_s(t)$ asymptotically, which implies that MAV velocities are estimated.□

It is noted that the MAV is a two-loop control system with required thrust inputting into the outer loop, and inner loop objective is Euler angle control. Attitude control frequency of the inner loop is higher than that of the outer loop. Usually, commercial MAVs directly receive Euler angles as inner loop inputs to realize attitude tracking.

We can, therefore, directly design the desired thrust and Euler angles required by the MAV, which are treated as control inputs to achieve the purpose of IBVS flight control. Specifically, MAV flight inputs that are thrust $U_1(t)$ and Euler angle vector $\boldsymbol{\eta}(t)$ can be calculated from the controller $\boldsymbol{f}(t)$ as follows [31]:

$$\begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix} = \frac{U_1}{m} \begin{bmatrix} \sin\theta_d \cos\varphi_d \\ -\sin\varphi_d \\ \cos\theta_d \cos\varphi_d \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}. \tag{49}$$

According to (49), under small angle approximation, the MAV thrust $U_1(t)$ can be obtained from $f_z(t)$ as

$$U_1 \approx m(f_z + g). \tag{50}$$

By using (49), the desired roll angle $\varphi_d(t)$ and pitch angle $\theta_d(t)$ can, moreover, be calculated as follows:

$$\theta_d = \arctan \frac{f_x}{f_z + g}, \quad \varphi_d = \arctan \frac{-f_y \cos\theta_d}{f_z + g}. \tag{51}$$

The yaw angular velocity of MAV is controlled using the proportional control law

$$u_\psi = k_4(\psi_d - \psi) \tag{52}$$

where $\psi_d(t)$ is the desired yaw angle.

During executing an IBVS task, it is noted that the adaptive law (32) does not satisfy the conditions of UUB, and therefore, it cannot guarantee the parameter $\hat{z}^*(t)$ to converge to truth value. The designed adaptive law, however, enables the controller act without requiring the desired depth z^*.

V. Simulation Results

In the section, the first simulation is conducted to verify feasibility of the proposed observer-based visual servoing controller. Then, the second simulation shows the results of the compared method [25].

Simulation 1: Feature point coordinates on the horizontal target plane are set under $\mathcal{F}^w$ initially as (0.25, 0.2, 0 m),

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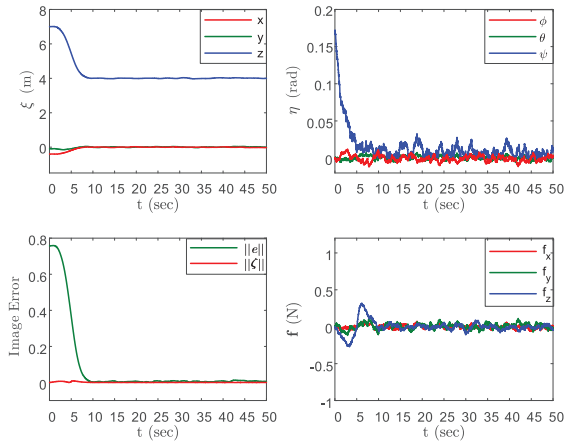

Fig. 1. Simulation 1: Variation of MAV states and system errors (top-left: MAV positions, top-right: MAV Euler angles, bottom-left: error vector norms on the virtual image plane, bottom-right: MAV resultant external forces).

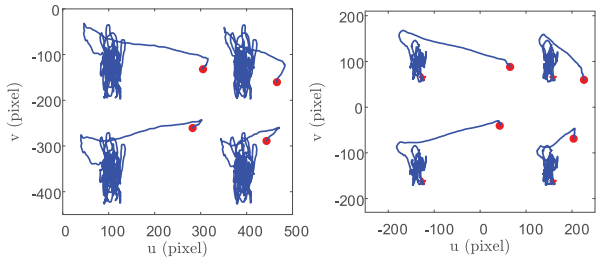

Fig. 2. Simulation 1: Paths of image features in the IBVS stage (left: on the real image plane, right: on the virtual image plane, circles: start pixels, stars: final pixels).

(0.25, −0.2, 0 m), (−0.25, 0.2, 0 m), and (−0.25, −0.2, 0 m), which are used to compute image moment features defined in (8). The camera has a focal length of 1.8 mm, and a sampling time is 0.01 s. MAV mass is 4.55 kg, and $g = 9.81$ m/s^2. Parameters $\rho_{\min}$ and $\rho_{\max}$ are considered to be 0.08 m^{-1} and 2 m^{-1}, respectively, corresponding to 12.5 and 0.5 m distances from the target, respectively. Controller and observer gains are chosen as $k_r = 8$, $k_1 = 0.2$, $k_2 = 20$, $k_3 = 2$, and $k_4 = 0.4$. The initial value of $\hat{z}^*(t)$ is set as $\hat{z}^*(0) = 10$ m for the adaptive law (32).

The MAV is set at a hovering position initially with the desired target in camera FOV, and the initial MAV hovering position is set as (−0.4, −0.1, 7 m) with the Euler angle as (0°, 0°, 10°) under $\mathcal{F}^w$. Desired image moments are set as $(q_x^d, q_y^d, q_z^d) = (0, 0, 1)$ with the parameter $a^* = 8.3 \times 10^{-8}$, implying that the MAV final position is (0, 0, 4 m) with zero Euler angles. White Gaussian noises of 5 dBW power are, moreover, added to target positions to simulate actual disturbance. By adding noises, the anti-interference ability of the system is tested with disturbed image data and target position.

Fig. 1 shows the variation of MAV states and error vectors, where the top-left displays the positions of the MAV, top-right shows Euler angles of the MAV, bottom-left provides the norms of image moment error and image moment estimation error on the virtual image plane, and the bottom-right shows the MAV resultant external forces. Fig. 2 gives paths of feature points on real and virtual image planes. Figs. 3 and 4 demonstrate the effectiveness of the proposed observers regarding image errors and MAV velocities, respectively. As shown in the two figures, the proposed observers make estimated signals converge to their truth values in about 5 s.

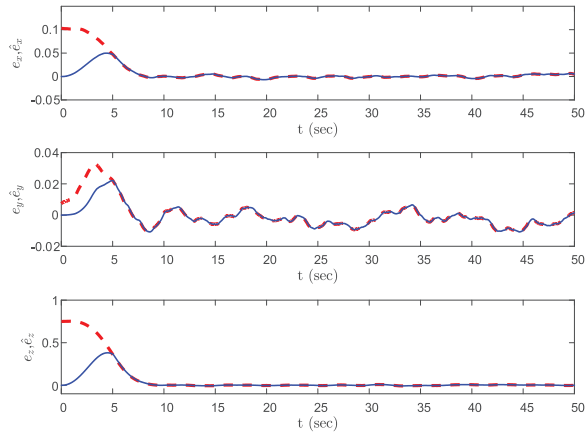

Fig. 3. Simulation 1: Observation $\hat{\boldsymbol{e}}(t)$ for the image moment error $\boldsymbol{e}(t)$ on the virtual image plane (solid lines: estimation values, dashed lines: truth values).

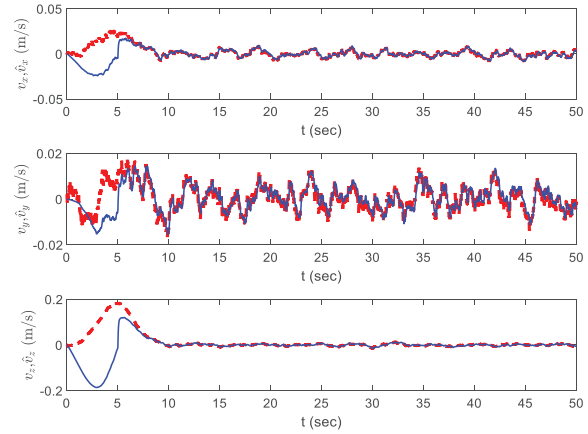

Fig. 4. Simulation 1: Observation for MAV velocities (solid lines: estimation values, dashed lines: truth values).

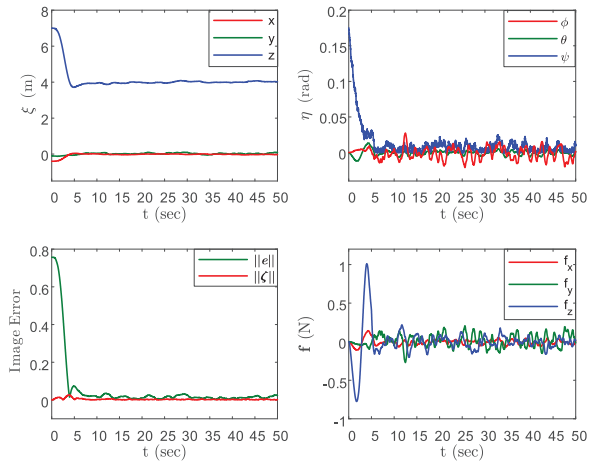

Fig. 5. Simulation 2: Variation of MAV states and system errors by [25] (top-left: MAV positions, top-right: MAV Euler angles, bottom-left: error vector norms on the virtual image plane, bottom-right: MAV resultant external forces).

Simulation 2: The method from [25] is implemented for comparison, and target parameters and initial conditions are kept identical to those in Simulation 1. Control gains are

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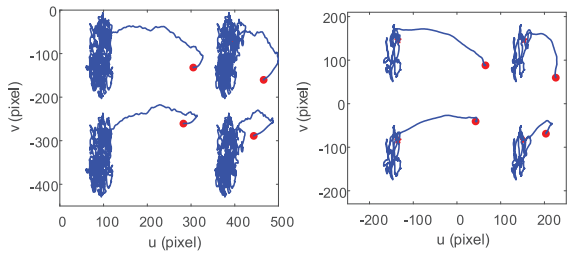

Fig. 6. Simulation 2: Paths of image features in the IBVS stage by [25] (left: on the real image plane, right: on the virtual image plane, circles: start pixels, stars: final pixels).

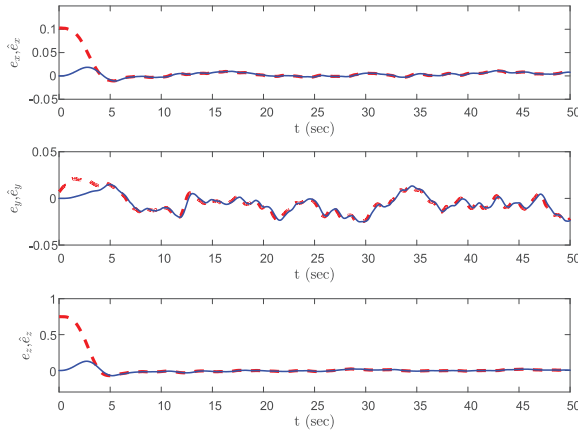

Fig. 7. Simulation 2: Observation for the image moment error on the virtual image plane by [25] (solid lines: estimated values, dashed lines: truth values).

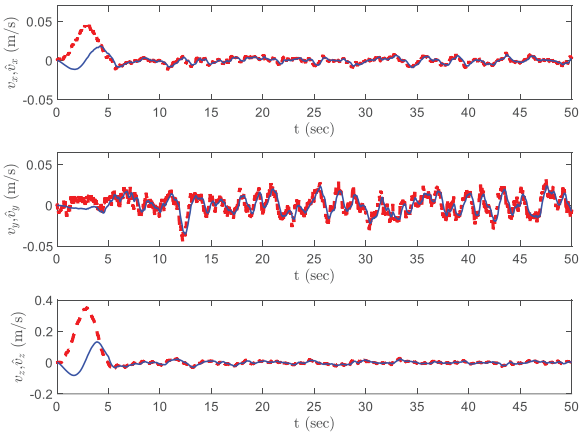

Fig. 8. Simulation 2: Observation for MAV velocities by [25] (solid lines: estimated values, dashed lines: truth values).

tuned as $k_r = 8$, $k_1 = 0.25, k_2 = 25$, and $k_4 = 0.4$ for good performance. Fig. 5 shows the evolution of MAV states and system errors, and Fig. 6 shows the paths of image features. Figs. 7 and 8 provide observation results for image moment error and MAV velocity, and it is seen that image feature errors and MAV velocities also converge to truth values.

Compared with [25], the proposed method converges slightly slower regarding MAV position errors but provides smaller control inputs and has less position fluctuation. The proposed method makes smoother image paths on both virtual and real image planes, which implies that the MAV flies steadily and its thrust and attitude are without sudden adjustment. The proposed method, moreover, makes more precise observations for image moment error and MAV velocity.

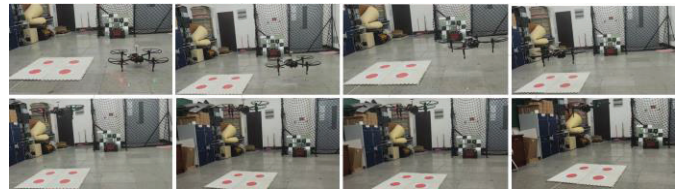

Fig. 9. Experiment 1: Several photos of the overall flight process.

VI. Experimental Results

In experiments, a DJI-M100 MAV equipped with dual cameras is used as the experimental platform, and a detailed structure is given in our previous work [32]. The experiments also consist of two stages of take-off and target search, which is presented in our previous work [24]. The proposed method and the compared method [25] are both treated as the IBVS stage. The forward camera is used to record MAV poses throughout the flight, and the downward camera is used during the IBVS stage. During the take-off and target search stages, the geometric controller [33] uses state information provided by the VINS-Mono algorithm [34] to control MAV flight, and then the MAV is controlled by the proposed controller in the IBVS stage. The first experiment is conducted to verify the effectiveness of the proposed controller, and the second experiment shows the results of the compared method [25].

Experiment 1: The target plane is placed on the ground, and the origin of the world frame is set at the centroid of four circles. The coordinates of circle centers are set arbitrarily as (0.25, 0.25, 0 m), (−0.25, 0.25, 0 m), (−0.25, −0.25, 0 m), and (0.25, −0.25, 0 m). The camera focal length is 1.8 mm. The desired image moments are set as $(q_x^d, q_y^d, q_z^d) = (0, 0, 1)$ with $a^* = 9.1 \times 10^{-7}$, which represents the MAV final position (0, 0, 1 m) with zero Euler angles under $\mathcal{F}^w$. MAV mass is 4.55 kg, and $g = 9.81$ m/s^2. Parameters $\rho_{\min}$ and $\rho_{\max}$ are set as 0.08 m^{-1} and 2 m^{-1}, respectively, corresponding to 12.5 and 0.5 m distances from the target, respectively. The sampling time of the closed-loop is 0.05 s and satisfies the real-time requirement.

Controller and observer gains are set as $k_r = 12.2$, $k_1 = 0.25$, $k_2 = 24.6$, $k_3 = 5.8$, and $k_4 = 0.4$. Initial value of $\hat{z}^*(t)$ is set as $\hat{z}^*(0) = 10$m for the adaptive update law (32). Initial flight position is set at (−1.2, 0.1, 0 m) with zero Euler angles under $\mathcal{F}^w$.

Several photos of the overall flight process are provided in Fig. 9. Fig. 10 shows the positions of the MAV and the proposed controller (33) makes the MAV stable at the desired position. The first 10 s are take-off and target search stages controlled by the MAV geometric controller [33]. Fig. 11 shows the control inputs of Euler angles and thrust for the MAV. Fig. 12 shows the norms of image moment error $\boldsymbol{e}(t)$ and image moment estimation error $\boldsymbol{\zeta}(t)$ on the virtual image plane. Fig. 13 provides the paths of feature points on real and virtual image planes in the IBVS stage, respectively.

The truth values for MAV velocities are obtained from the VINS-Mono strategy [34] for evaluating observation accuracy.

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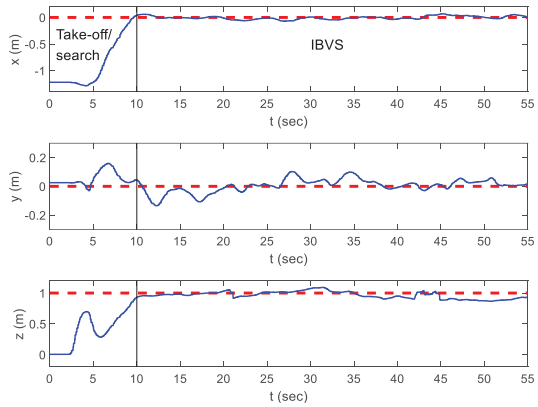

Fig. 10. Experiment 1: Evolution of MAV positions in overall flight (dashed lines: desired positions, solid lines: current positions).

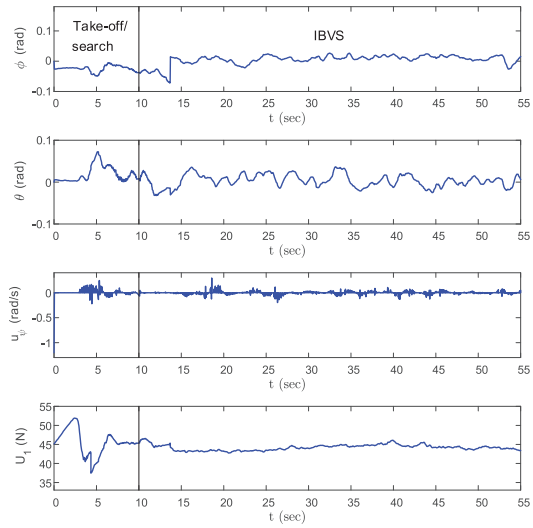

Fig. 11. Experiment 1: Control inputs of Euler angles and thrust for the MAV.

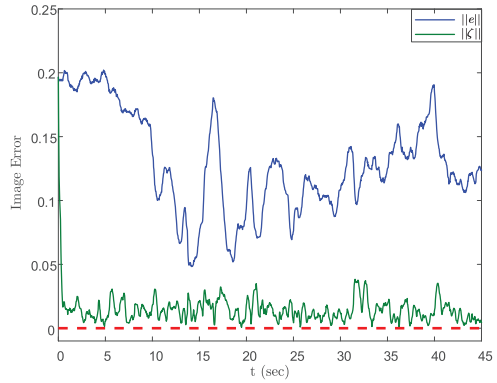

Fig. 12. Experiment 1: Norms of image error vectors on the virtual image plane (dashed line: expected zero values, solid lines: current values of error vector norms).

Figs. 14 and 15 represent the outputs of the proposed observers for image moment error and MAV velocity, respectively. As shown in the two figures, the proposed observers make estimated signals converge to their truth values precisely in a short period of time.

Experiment 2: The method proposed in [25] is conducted as a comparison, and target parameters and initial conditions are kept identical to those in Experiment 1. Control gains are

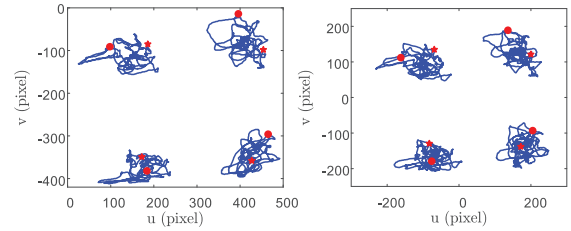

Fig. 13. Experiment 1: Paths of image features in the IBVS stage (left: on the image plane, right: on the virtual image plane, circles: start pixels, stars: final pixels).

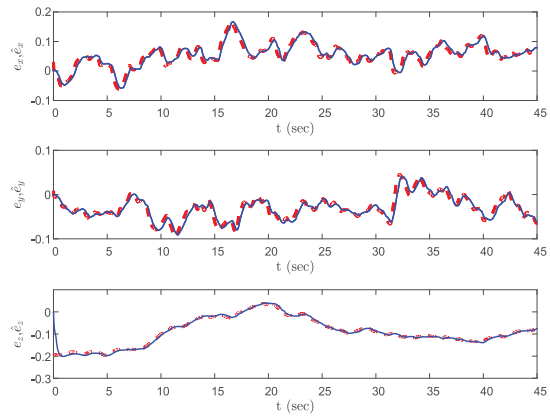

Fig. 14. Experiment 1: Observation for the image moment error on the virtual image plane (solid lines: estimated values, dashed lines: truth values).

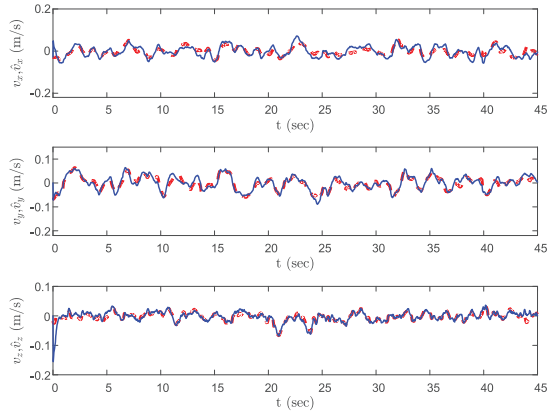

Fig. 15. Experiment 1: Observation for MAV velocities (solid lines: estimated values, dashed lines: truth values).

selected as $k_r = 15$, $k_1 = 0.4$, $k_2 = 25$, and $k_4 = 0.4$ for good performance.

Fig. 16 shows positions of the MAV, and the IBVS stage is conducted by [25]. Fig. 17 shows control inputs of Euler angles and thrust for the MAV. It is seen that the thrust by the proposed method is smaller than that by the compared method, and the yaw angular velocity input is less chattering with the proposed method. Figs. 18 and 19 represent observation for image moment error and MAV velocity, respectively.

Consequently, depth adaptation is incorporated in design of the proposed IBVS controller, which improves robustness of the visual servoing controller in unknown depth environments. From experimental results, it can be seen that the proposed method makes the MAV hover stably at the desired position

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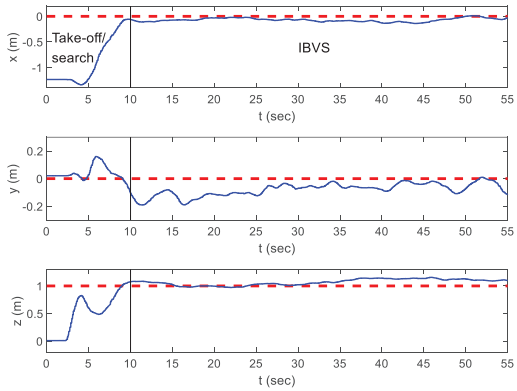

Fig. 16. Experiment 2: Evolution of MAV positions in overall flight by 0 (dashed lines: desired positions, solid lines: actual positions).

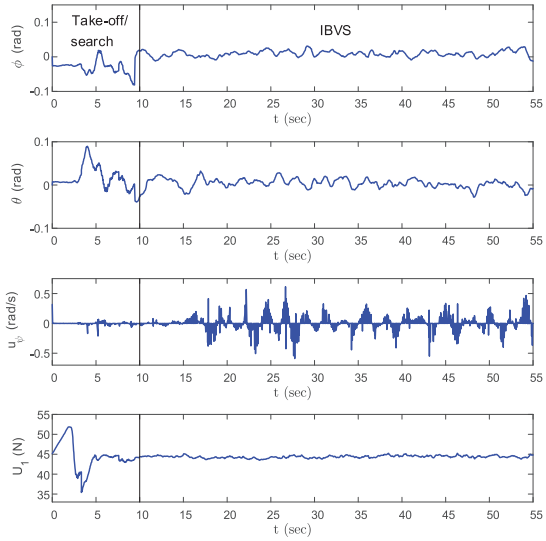

Fig. 17. Experiment 2: Control inputs of Euler angles and thrust for the MAV by [25].

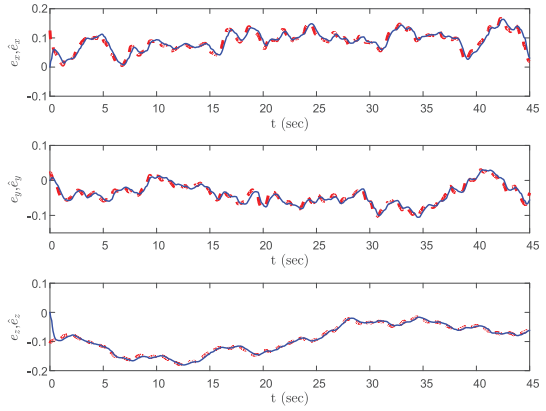

Fig. 18. Experiment 2: Observation for the image moment error on the virtual image plane by [25] (solid lines: estimated values, dashed lines: truth values).

and attitude, and the proposed observers make more accurate estimation for the MAV velocity.

VII. Conclusion

In the article, an output feedback IBVS method is proposed with state observers for controlling quadrotor MAV motion. By defining appropriate image moment features on the virtual image plane, it aims to obtain satisfied regulation trajectories in both image and Cartesian space. IMUs commonly used in MAV motion do not provide accurate linear velocity information and suffer from accumulated errors. The flow of image features is, thus, used as the linear velocity cue in the article, which is estimated from selected visual features by designing nonlinear observers. An adaptive law is designed for unknown image depth information, which is combined with observers in output feedback IBVS controller design to enhance system robustness. Stability analysis shows that all states of the system are bounded, and the error signals converge to zero. Simulation and experimental results show that the proposed method responds well in both image space and Cartesian space and provides accurate estimation for image moment errors and MAV velocities. In the future, linear velocity observers will be designed regarding both MAV and dynamic targets so as to achieve image-based visual servo tracking tasks with robustness.

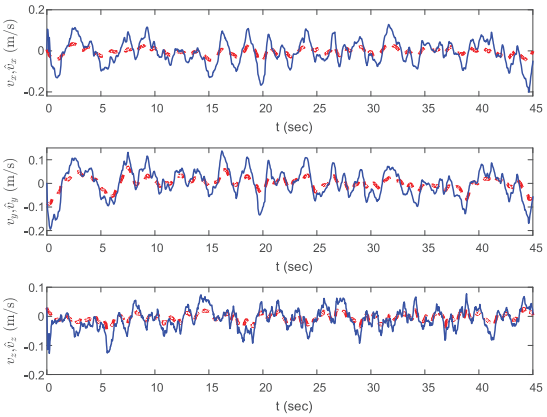

Fig. 19. Experiment 2: Observation for MAV velocities by [25] (solid lines: estimated values, dashed lines: truth values).

References

[1] J. Wu et al., "An adaptive conversion speed Q-learning algorithm for search and rescue UAV path planning in unknown environments," *IEEE Trans. Veh. Technol.*, vol. 72, no. 12, pp. 15391–15404, Dec. 2023.

[2] T. Yang, Z. Jiang, R. Sun, N. Cheng, and H. Feng, "Maritime search and rescue based on group mobile computing for unmanned aerial vehicles and unmanned surface vehicles," *IEEE Trans. Ind. Informat.*, vol. 16, no. 12, pp. 7700–7708, Dec. 2020.

[3] H. Xie, A. F. Lynch, K. H. Low, and S. Mao, "Adaptive output-feedback image-based visual servoing for quadrotor unmanned aerial vehicles," *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 3, pp. 1034–1041, May 2020.

[4] W. Xu, D. He, Y. Cai, and F. Zhang, "Robots' state estimation and observability analysis based on statistical motion models," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 5, pp. 2030–2045, Sep. 2022.

[5] J. Thomas, J. Welde, G. Loianno, K. Daniilidis, and V. Kumar, "Autonomous flight for detection, localization, and tracking of moving targets with a small quadrotor," *IEEE Robot. Autom. Lett.*, vol. 2, no. 3, pp. 1762–1769, Jul. 2017.

[6] R. Mebarki, V. Lippiello, and B. Siciliano, "Nonlinear visual control of unmanned aerial vehicles in GPS-denied environments," *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 1004–1017, Aug. 2015.

[7] J. Zheng, T. Yang, H. Liu, T. Su, and L. Wan, "Accurate detection and localization of unmanned aerial vehicle swarms-enabled mobile edge computing system," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 5059–5067, Jul. 2021.

[8] J. Ye, C. Fu, F. Lin, F. Ding, S. An, and G. Lu, "Multi-regularized correlation filter for UAV tracking and self-localization," *IEEE Trans. Ind. Electron.*, vol. 69, no. 6, pp. 6004–6014, Jun. 2022.

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[9] J. Lin, Z. Miao, Y. Wang, H. Wang, X. Wang, and R. Fierro, "Vision-based safety-critical landing control of quadrotors with external uncertainties and collision avoidance," *IEEE Trans. Control Syst. Technol.*, vol. 32, no. 4, pp. 1310–1322, Jul. 2024.

[10] L. Xu, T. Wang, J. Wang, J. Liu, and C. Sun, "Attention-based policy distillation for UAV simultaneous target tracking and obstacle avoidance," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 2, pp. 3768–3781, Feb. 2024.

[11] L. Yu, E. Yang, B. Yang, Z. Fei, and C. Niu, "A robust learned feature-based visual odometry system for UAV pose estimation in challenging indoor environments," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.

[12] L. Teixeira, F. Maffra, M. Moos, and M. Chli, "VI-RPE: Visual-inertial relative pose estimation for aerial vehicles," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 2770–2777, Oct. 2018.

[13] Y. Chen, S. Huang, and R. Fitch, "Active SLAM for mobile robots with area coverage and obstacle avoidance," *IEEE/ASME Trans. Mechatronics*, vol. 25, no. 3, pp. 1182–1192, Jun. 2020.

[14] F. Romanelli, F. Martinelli, and E. D. Giampaolo, "Robust simultaneous localization and mapping using range and bearing estimation of radio ultra high frequency identification tags," *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 2, pp. 772–785, Mar. 2023.

[15] Z. Wang, X. Li, Y. Zhang, and P. Huang, "Localization, planning, and control of a UAV for rapid complete coverage bridge inspection in large-scale intermittent GPS environments," *IEEE Trans. Control Syst. Technol.*, vol. 32, no. 4, pp. 1357–1369, Jul. 2024.

[16] X. Zhang, Y. Fang, X. Zhang, J. Jiang, and X. Chen, "Dynamic image-based output feedback control for visual servoing of multirotors," *IEEE Trans. Ind. Informat.*, vol. 16, no. 12, pp. 7624–7636, Dec. 2020.

[17] H. Jabbari, G. Oriolo, and H. Bolandi, "An adaptive scheme for image-based visual servoing of an underactuated UAV," *Int. J. Robot. Autom.*, vol. 29, no. 1, pp. 92–104, May 2014.

[18] A. Yesildirek and B. Imran, "Nonlinear control of quadrotor using multi Lyapunov functions," in *Proc. Amer. Control Conf.*, Portland, OR, USA, Jun. 2014, pp. 3844–3849.

[19] R. Ozawa and F. Chaumette, "Dynamic visual servoing with image moments for an unmanned aerial vehicle using a virtual spring approach," *Adv. Robot.*, vol. 27, no. 9, pp. 683–696, Jun. 2013.

[20] H. de Plinval, P. Morin, P. Mouyon, and T. Hamel, "Visual servoing for underactuated VTOL UAVs: A linear, homography-based framework," *Int. J. Robust Nonlinear Control*, vol. 24, no. 16, pp. 2285–2308, Nov. 2014.

[21] H. Xie and A. F. Lynch, "Input saturated visual servoing for unmanned aerial vehicles," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 2, pp. 952–960, Apr. 2017.

[22] K. Zhang, Y. Shi, and H. Sheng, "Robust nonlinear model predictive control based visual servoing of quadrotor UAVs," *IEEE/ASME Trans. Mechatronics*, vol. 26, no. 2, pp. 700–708, Apr. 2021.

[23] Y. Huang, M. Zhu, T. Chen, and Z. Zheng, "Robust homography-based visual servo control for a quadrotor UAV tracking a moving target," *J. Franklin Inst.*, vol. 360, no. 3, pp. 1953–1977, Feb. 2023.

[24] L. Shi, B. Li, W. Shi, and X. Zhang, "Visual servoing of quadrotor UAVs for slant targets with autonomous object search," *Proc. Inst. Mech. Eng., I, J. Syst. Control Eng.*, vol. 237, no. 5, pp. 791–804, Jan. 2023.

[25] H. J. Asl and J. Yoon, "Vision-based control of a flying robot without linear velocity measurements," in *Proc. IEEE Int. Conf. Adv. Intell. Mechatronics (AIM)*, Jul. 2015, pp. 1670–1675.

[26] G. Fink, H. Xie, A. F. Lynch, and M. Jagersand, "Experimental validation of dynamic visual servoing for a quadrotor using a virtual camera," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Denver, CO, USA, Jun. 2015, pp. 1231–1240.

[27] H. A. Hashim, "Exponentially stable observer-based controller for VTOL-UAVs without velocity measurements," *Int. J. Control*, vol. 96, no. 8, pp. 1946–1960, Aug. 2023.

[28] H. J. Asl, G. Oriolo, and H. Bolandi, "Output feedback image-based visual servoing control of an underactuated unmanned aerial vehicle," *Proc. Inst. Mech. Eng. I, J. Syst. Control Eng.*, vol. 228, no. 7, pp. 435–448, Aug. 2014.

[29] J. Gao, A. A. Proctor, Y. Shi, and C. Bradley, "Hierarchical model predictive image-based visual servoing of underwater vehicles with adaptive neural network dynamic control," *IEEE Trans. Cybern.*, vol. 46, no. 10, pp. 2323–2334, Oct. 2016.

[30] D. Zheng, H. Wang, J. Wang, S. Chen, W. Chen, and X. Liang, "Image-based visual servoing of a quadrotor using virtual camera approach," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 2, pp. 972–982, Apr. 2017.

[31] L. Shi, B. Li, and W. Shi, "Vision-based UAV adaptive tracking control for moving targets with velocity observation," *Trans. Inst. Meas. Control*, vol. 46, no. 12, pp. 2255–2265, Aug. 2024.

[32] C. An, B. Li, W. Shi, and X. Zhang, "Autonomous quadrotor UAV systems for dynamic platform landing with onboard sensors," *Int. J. Robot. Autom.*, vol. 38, no. 4, pp. 296–305, 2023.

[33] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *Proc. 49th IEEE Conf. Decis. Control (CDC)*, Atlanta, GA, USA, Dec. 2010, pp. 5420–5425.

[34] T. Qin, P. Li, and S. Shen, "VINS-mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 1004–1020, Aug. 2018.

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